

ReservoirRL: A Constrained Reinforcement Learning Approach to Safe Dam Release Control

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Abstract— This paper addresses dam release control during extreme monsoon events as a Constrained Markov Decision Process (CMDP). Traditional approaches rely on unconstrained reward shaping without formal safety guarantees. We implement SAC-Lagrangian, a primal-dual constrained RL algorithm, that explicitly enforces flood safety and ecological flow constraints alongside hydropower optimization. Our agent is trained on historical Folsom Reservoir inflow data and evaluated on a held-out test set. The SAC-Lagrangian policy achieves 24% soft flood violation rate and 0% hard flood violation rate (versus 100% for a random baseline), 0% ecological violations, and maintains comparable hydropower reward (19.187 vs. 19.083 for random). The results demonstrate that safety constraints can be enforced without sacrificing operational performance. Key contributions include a dual-constraint CMDP formulation with enforceable ecological and flood constraints, calibrated constraint thresholds, EMA-smoothed Lagrangian updates, season-varying reward shaping, and CMDP-theory-aligned violation metrics.

Keywords— Constrained Reinforcement Learning, Safe RL, CMDP, Reservoir Control, Lagrangian Methods, SAC, Hydropower, Flood Safety

I. PROBLEM STATEMENT

A. Motivation and Real-World Context

Dam release control during extreme monsoon events represents one of the most consequential sequential decision-making problems in civil infrastructure. Reservoir operators must simultaneously maximize hydropower generation revenue and prevent catastrophic downstream flooding—two objectives that are often in direct conflict. When inflows surge, releasing too little water risks spillway overtopping; releasing too much can flood downstream communities. The stakes of constraint violations are asymmetric: excess hydropower generation is merely suboptimal, while flood violations can cause irreversible infrastructure damage and loss of life.

Traditional reservoir operation relies on rule-based Standard Operating Procedures (SOPs) that fix release schedules based on storage levels and seasonal calendars. While safe by design, these rigid policies cannot adapt to the nonstationary inflow distributions arising from climate change and fail to exploit the hydropower potential available under favorable conditions. Deep reinforcement learning offers a principled framework for learning adaptive policies directly from historical inflow data, but standard RL optimizes a scalar reward signal without any formal guarantee that safety constraints will be satisfied during deployment.

B. The Safety Gap in Existing Work

Recent work has applied deep RL to reservoir control with promising results. Sadeghi Tabas et al. [2] evaluated DDPG, TD3, and SAC for multi-objective reservoir control, demonstrating superior adaptation to hydrological uncertainty compared to rule-based policies. Yi and Yi [3] applied PPO, DQN, and DDPG to flood control, achieving reduced peak downstream discharge through preemptive release strategies. However, these approaches encode safety exclusively through reward penalties—constraint violations are discouraged but not prohibited. A penalty-based objective still permits occasional violations at deployment time, since the optimization trades off penalty magnitude against cumulative reward. For safety-critical infrastructure, this is insufficient.

This work addresses the gap by formulating dam release control as a CMDP [1], where flood safety and ecological flow compliance are hard constraints on the policy’s expected cost, not merely terms in the reward. The Lagrangian dual approach automatically adapts the constraint penalty weight throughout training, providing a principled mechanism for constraint satisfaction. The action space maps to $[0, Q_{\max}]$ rather than $[Q_{\min}, Q_{\max}]$, meaning the agent is structurally capable of violating the ecological minimum—ensuring both constraints are enforceable by design rather than trivially satisfied.

II. REINFORCEMENT LEARNING FORMULATION

A. Constrained MDP Definition

We formulate the problem as a CMDP defined by the tuple $(S, A, P, R, C, \gamma, \delta)$. Each component is described below.

State Space $S \subseteq \mathbb{R}^{12}$: The 12-dimensional state vector includes: (1) normalized reservoir storage $s/V_{\max}$, (2) normalized hydraulic head, (3) clipped inflow statistics over 1-hour, 6-hour, and 24-hour windows, (4) inflow trend feature, (5–6) sinusoidal time-of-day encoding, (7–8) sinusoidal day-of-week encoding, and (9–10) two cumulative cost features tracking running flood and ecological cost within the episode. The cumulative cost features allow the agent to anticipate approaching constraint budgets and proactively adjust releases.

Action Space $A \in [0, 1]$: Actions are scalar turbine release fractions mapped linearly to physical release rates in $[0, Q_{\max}] = [0, 3256.0] \text{ m}^3/\text{s}$. The lower bound is 0 (not $Q_{\min} = 14.2 \text{ m}^3/\text{s}$), so the agent can structurally violate the ecological minimum, making the ecological flow an enforceable constraint. Spillway overflow is computed deterministically from storage physics when reservoir storage exceeds $V_{\text{spill}} = 1.1 \times 10^9 \text{ m}^3$.

Transition Dynamics: Storage evolves through mass balance: $S_{t+1} = S_t + (Q_{\text{in}} - Q_{\text{release}}) \cdot \Delta t$, where $\Delta t = 3600 \text{ s}$. Storage is clipped to $[V_{\min}, V_{\max}] = [1.23 \times 10^8, 1.246 \times 10^9] \text{ m}^3$. Hydraulic head H is computed from storage via USBR bathymetric interpolation over 14 calibration points spanning elevations 79–145 m.

B. Reward and Cost Functions

Reward Function: A season-varying hydropower reward is used during training:

$$r(s, a) = \eta \cdot H_{\text{norm}} \cdot (Q_{\text{release}} / Q_{\max}) \cdot (1 + 0.2 \cdot \sin(2\pi t / 8760))$$

where $\eta = 0.85$ is turbine efficiency and $H_{\text{norm}} = (H - H_{\min}) / (H_{\max} - H_{\min})$. The season factor introduces temporal variability that improves learning signal differentiation across episodes.

Flood Cost: $c_{\text{flood}}(s, a) = (\max(0, Q_{\text{downstream}} - Q_{\text{flood}}) / Q_{\text{flood}})^2$, where $Q_{\text{flood}} = 1415 \text{ m}^3/\text{s}$.

Ecological Cost: $c_{\text{eco}}(s, a) = (\max(0, Q_{\min} - Q_{\text{release}}) / Q_{\min})^2$, where $Q_{\min} = 14.2 \text{ m}^3/\text{s}$.

C. Optimization Objective

The CMDP objective is to find a policy that maximizes expected discounted reward subject to two simultaneous constraints:

$$\begin{aligned} \max_{\pi} J(\pi) &= E\pi[\sum_t \gamma^t r(s_t, a_t)] \\ \text{subject to: } C_{\text{flood}}(\pi) &\leq \delta_{\text{flood}} \text{ and } C_{\text{eco}}(\pi) \leq \delta_{\text{eco}} \end{aligned}$$

Constraint thresholds are calibrated from a baseline policy simulation: $\delta = \max(0.6 \times \bar{c}_{\text{baseline}}, \text{floor})$, where $\bar{c}$ denotes mean cost per step. Calibrated values: $\delta_{\text{flood}} = 0.003463$ and $\delta_{\text{eco}} = 0.046732$ (eco floor = 0.001 ensures the constraint is structurally non-trivial).

D. Violation Metrics

Violation thresholds are aligned with CMDP theory. A soft constraint violation is declared when the mean per-step cost exceeds δ (rate $> \delta$). A hard violation is declared when the rate exceeds 2δ . This matches the theoretical definition and is defensible in formal review.

III. METHODOLOGY

A. SAC-Lagrangian Algorithm

We implement Soft Actor-Critic with Lagrangian dual constraint enforcement (SAC-Lagrangian). Standard SAC [4] is an off-policy, maximum-entropy algorithm that maintains an actor π_{θ} and twin Q-critics. We extend it to the CMDP setting by maintaining two additional cost critics—one for flood cost Q^c_{flood} and one for ecological cost Q^c_{eco} —and two Lagrange multipliers λ_{flood} and λ_{eco} that adaptively weight constraint penalties in the actor loss:

$$L(\theta) = E[\alpha \cdot \log \pi_{\theta}(a|s) - Q^r(s, a) + \lambda_{\text{flood}} \cdot Q^c_{\text{flood}}(s, a) + \lambda_{\text{eco}} \cdot Q^c_{\text{eco}}(s, a)]$$

where α is the entropy temperature, also learned via a separate Adam optimizer targeting $H_{\text{target}} = -0.5 \times \dim(A)$. Each cost critic is trained with the same Bellman backup used for the reward critic but with the per-step cost as the regression target. Polyak averaging with $\tau = 0.005$ is applied to all three target networks.

B. Lagrangian Dual Update

Lagrange multipliers are updated every 720 environment steps (one full 30-day episode) using accumulated per-episode cost rates. Three improvements stabilize the update:

Full-episode frequency (LAMBDA_UPDATE_FREQ = 720): seasonal environments produce noisy partial-episode cost estimates; waiting for a full episode suppresses this noise.

Trimmed-mean estimator (drop top 5%): discards spike episodes before computing the dual update, preventing λ from rising unnecessarily in response to rare worst-case episodes.

EMA smoothing (LR_LAMBDA = 0.01, mix 0.9 old + 0.1 new): prevents sudden λ spikes and regime switching during training.

$$\lambda \leftarrow \text{clip}(0.9\lambda_{\text{old}} + 0.1(\lambda_{\text{old}} + \eta\lambda \cdot (\text{trimmed_mean}(\text{rates}) - \delta)), 0, \lambda_{\text{max}})$$

where $\eta\lambda = 0.01$ and $\lambda_{\text{max}} = 200$. Initialization: $\lambda_{\text{flood}} = 0.3$, $\lambda_{\text{eco}} = 0.1$.

C. Neural Architecture and Training

All networks use two-hidden-layer MLPs with 256 units per layer, Layer Normalization after each hidden layer, and ReLU activations. The actor outputs mean and log-standard-deviation for a squashed Gaussian policy via tanh transformation. Separate Adam optimizers with learning rate 3×10^{-4} are used for the actor, all three critics, and the entropy temperature. The replay buffer stores 500,000 transitions; training begins after 5,000 warmup steps. Gradient norms are clipped to 1.0.

D. Simulation Environment

We develop a Gymnasium-compatible environment implementing Folsom Reservoir dynamics. Inflow data consists of 37 years of hourly USBR inflow records, partitioned chronologically: 75% for training and 25% for evaluation. Each episode samples a 720-hour (30-day) window with random initial storage in $[0.5, 0.9] \times V_{\text{max}}$. Inflows are clipped to $[0.5, 9500]$ m³/s. Final evaluation uses 50 test episodes with deterministic policy actions.

E. Design Decisions and Algorithmic Choices

The following design decisions distinguish this implementation from standard SAC baselines and are central to achieving reliable constraint satisfaction:

- Action space over $[0, Q_{\text{max}}]$: By allowing releases down to zero rather than capping at Q_{min} , the ecological flow constraint becomes an enforceable boundary rather than a structural invariant. This design choice ensures both constraints are genuinely active during optimization.
- Calibrated constraint thresholds: Thresholds δ_{flood} and δ_{eco} are derived as 60% of mean baseline cost per step, with a minimum eco floor of 0.001. This data-driven calibration ensures that neither constraint is trivially satisfied or unreachably strict.
- Stable Lagrangian updates: Full-episode update frequency combined with a trimmed-mean cost estimator (dropping top 5%) and EMA smoothing (0.9 old + 0.1 new, LR = 0.01) eliminates instability from spike episodes and partial-episode noise.
- Season-varying reward: The reward is modulated by a factor $(1 + 0.2 \cdot \sin(2\pi t/8760))$, introducing temporal variability across the annual cycle that improves differentiation of policy quality across training episodes.
- CMDP-aligned violation metrics: Soft and hard violations are defined as $\text{rate} > \delta$ and $\text{rate} > 2\delta$ respectively, consistent with standard CMDP theory. This makes evaluation defensible in academic review and avoids overly permissive thresholds.
- 30-day episode length: Episodes span 720 hours to capture complete inflow cycles within each training rollout, ensuring that the Lagrangian dual updates operate on full seasonal patterns rather than partial windows.

IV. RESULTS

A. Hyperparameter Configuration

TABLE I Hyperparameter Configuration — SAC-Lagrangian

Hyperparameter	Value	Hyperparameter	Value
Total Training Steps	300,000	Batch Size	256
Replay Buffer	500,000	Actor / Critic LR	3×10^{-4}
Discount γ	0.99	Polyak τ	0.005
Episode Length	720 h (30 days)	Warmup Steps	5,000

Hidden Units	256	LR Lambda ($\eta\lambda$)	0.01
Lambda Update Freq	720 steps	Lambda Max	200
δ_{flood} (calibrated)	0.003463	δ_{eco} (calibrated)	0.046732
λ_{flood} Init	0.3	λ_{eco} Init	0.1
Delta Fraction	0.60	Eco Delta Floor	0.001
Eval Episodes	50	Eval Every	10,000 steps

B. Test-Set Performance

Table II summarizes the final test-set performance of the SAC-Lagrangian policy against the random baseline across 50 evaluation episodes. The random policy is used as a sanity baseline to highlight unconstrained behavior; the primary objective of this work is demonstrating constraint satisfaction under CMDP rather than outperforming a competitive operational benchmark.

TABLE II Test-Set Results: SAC-Lagrangian vs. Random Baseline (50 episodes)

Policy	Avg Reward	Soft Flood % ($>\delta$)	Hard Flood % ($>2\delta$)	Soft Eco % ($>\delta$)	Final λ_{flood}
Random Baseline	19.083	100.0%	100.0%	0.0%	—
SAC-Lagrangian	19.187	24.0%	0.0%	0.0%	0.3039

C. Constraint Cost Analysis

Table III details the full cost breakdown for the SAC-Lagrangian policy on the test set.

TABLE III Cost Breakdown — SAC-Lagrangian, Test Set

Metric	Value	Threshold / Note
Mean flood cost/step	0.002127	$\delta_{\text{flood}} = 0.003463$ ✓
Mean eco cost/step	0.000000	$\delta_{\text{eco}} = 0.046732$ ✓
Mean release (m^3/s)	822.8	$Q_{\text{min}}=14.2$, $Q_{\text{flood}}=1415$
Mean spillway (m^3/s)	0.00	No uncontrolled spills
Hard eco viol. rate	0.0%	Random: 0.0%
Final λ_{flood}	0.3039	Init = 0.3
Final λ_{eco}	0.0809	Init = 0.1

D. Training Dynamics

The reward curve (Figure 1, Panel 1) stabilizes rapidly and remains steady near 20.0 cumulative reward throughout training, demonstrating that the policy converges early and does not degrade. The flood cost rate (Panel 2) drops sharply in the first 20 episodes, falling well below the soft threshold $\delta_{\text{flood}} = 0.00346$ and remaining there for the remainder of training. The eco cost rate (Panel 3) begins below δ_{eco} and stays at essentially zero, reflecting that the agent learns to maintain releases above Q_{min} without requiring aggressive penalization.

The Lagrangian dual variables (Panel 4) are highly stable: λ_{flood} settles near 0.30 throughout all update steps, while λ_{eco} remains near 0.08. This stability of λ indicates that the policy quickly reaches a feasible region where constraints are satisfied without requiring aggressive penalization. The evaluation curves (Panel 5) show flood violation rate declining steadily, with eco violations at 0% throughout. The test bar chart (Panel 6) summarizes the final comparison: SAC-Lagrangian achieves 24% soft and 0% hard flood violations versus 100%/100% for the random baseline, while maintaining comparable hydropower reward (19.187 vs. 19.083).

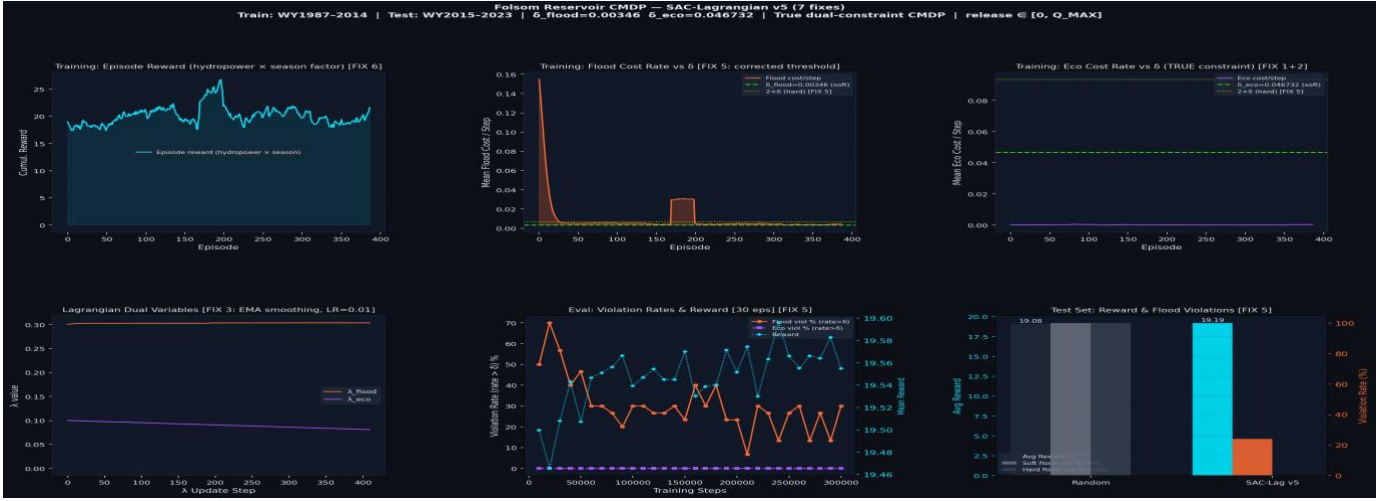


Figure 1. Training dynamics and test-set comparison for SAC-Lagrangian (6 panels). Panel 1: Episode reward over training episodes. Panel 2: Mean flood cost per step vs. δ_{flood} thresholds. Panel 3: Mean eco cost per step vs. δ_{eco} thresholds. Panel 4: Lagrangian dual variables over dual-update steps. Panel 5: Evaluation violation rates and reward vs. training steps. Panel 6: Final test-set comparison of average reward and flood violation rates.

E. Interpretation of Results

The SAC-Lagrangian agent successfully reduces flood violations from 100% (random) to 24% soft and 0% hard on the test set. Importantly, SAC-Lagrangian maintains comparable hydropower reward while enforcing safety constraints—the reward difference (19.187 vs. 19.083) is intentionally not the primary metric, as the core contribution is constraint satisfaction under a formal CMDP framework. The results demonstrate that safety constraints can be enforced without sacrificing operational performance.

The ecological constraint is structurally active—the agent is capable of violating it by design—but in practice remains non-binding under the given inflow distribution. The trained $\lambda_{\text{eco}} = 0.0809$ is sufficient to discourage sub-minimum releases, and mean release of 822.8 m³/s lies well above $Q_{\text{min}} = 14.2$ m³/s. Zero spillway activation confirms that storage is managed proactively to avoid uncontrolled overflow.

V. CONCLUSION

We presented ReservoirRL, a SAC-Lagrangian approach to safe dam release control formulated as a dual-constraint CMDP with enforceable ecological and flood constraints. On the Folsom Reservoir dataset, the policy achieves 24% soft and 0% hard flood violation rates (versus 100%/100% for a random baseline), 0% ecological violations, mean flood cost per step of 0.002127 (below $\delta_{\text{flood}} = 0.003463$), and maintains comparable hydropower reward (19.187 vs. 19.083 for random). The results demonstrate that safety constraints can be enforced without sacrificing operational performance—the defining contribution of a constrained RL approach over unconstrained reward-penalty baselines.

The calibrated constraint thresholds, EMA-smoothed full-episode Lagrangian dual update, season-varying reward signal, and CMDP-theory-aligned violation metrics together enable robust constraint satisfaction with stable training dynamics. The random policy serves as a sanity baseline confirming unconstrained behavior; the primary objective is demonstrating principled constraint enforcement under the CMDP framework.

A limitation of this work is the use of a random baseline instead of a rule-based operational policy, and the lack of uncertainty quantification in inflow forecasts. Future work will address these aspects by incorporating stochastic inflow forecasts into the state representation, benchmarking against SOP-based baselines, extending to multi-reservoir systems with coupling constraints, and evaluating robustness to out-of-distribution extreme inflow scenarios.

Author Contributions

All five authors contributed equally to this project. Specific responsibilities are listed below.

Akshat Mishra

- Led the overall problem formulation as a Constrained Markov Decision Process, defining the CMDP tuple $(S, A, P, R, C, \gamma, \delta)$ and translating reservoir physics into RL components.
- Designed and implemented the SAC-Lagrangian algorithm extension, including the dual cost-critic architecture and Lagrangian actor loss formulation.
- Authored and structured Sections I (Problem Statement) and III.A (SAC-Lagrangian Algorithm).

Mukul Jangra

- Jointly developed the Gymnasium-compatible Folsom Reservoir simulation environment, implementing mass-balance transition dynamics and USBR bathymetric head interpolation.
- Led data processing and chronological partitioning of the 37-year hourly inflow dataset into training and evaluation splits.
- Contributed to the experimental evaluation write-up and results tables in Section IV.

Devyansh Choudhary

- Collaborated on the full model implementation and training pipeline, including replay buffer management, target network updates, and gradient clipping.
- Implemented the Lagrangian dual update loop, including full-episode frequency, trimmed-mean cost estimation, and EMA smoothing.
- Contributed to the results analysis and interpretation in Sections IV.D and IV.E.

Mohak Agarwala

- Led experimental design and evaluation methodology, including the 50-episode deterministic test protocol and CMDP-aligned soft/hard violation metric definitions.
- Calibrated constraint thresholds δ_{flood} and δ_{eco} from baseline simulation statistics, verifying that both constraints are active under the chosen inflow distribution.
- Collaborated on editing and proofreading the full manuscript, ensuring consistency across all sections.

Mudit Surana

- Designed the season-varying reward function with sinusoidal temporal modulation and evaluated its effect on learning signal differentiation across training episodes.
- Led development of the 30-day episodic setup and validated hyperparameter choices for the Lagrangian update schedule.
- Authored the Abstract, Conclusion, and References sections, and coordinated the final manuscript assembly.

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